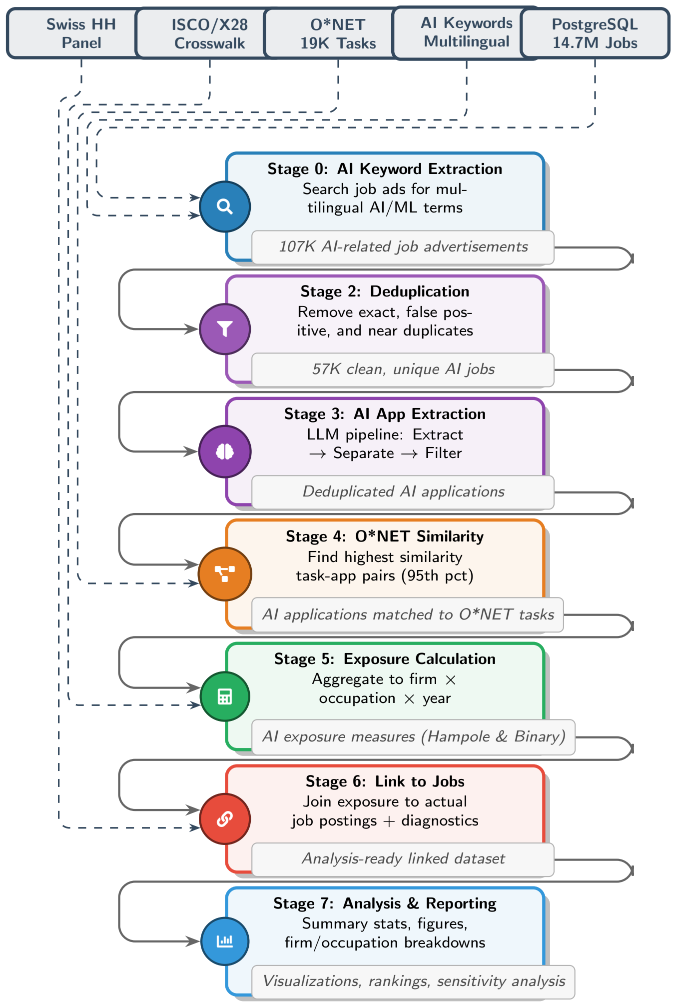

Figure 1: AI Exposure Pipeline: From Job Advertisements to Individual-Level Analysis

Pipeline Summary

Data Inputs

- **PostgreSQL**: 14.7M job advertisements from Swiss job market
- **AI Keywords**: Multilingual keyword list (EN, DE, FR, IT)
- **O*NET**: 19K occupational tasks with importance weights
- **ISCO/X28 Crosswalk**: Occupational classification mapping
- **Swiss Household Panel**: Individual respondent data for exposure linkage

Stage Descriptions

Stage 0: AI Keyword Extraction
Uses multilingual keyword matching to identify AI-related job postings. Reduces 14.7M jobs to approximately 107K AI-relevant advertisements.

Stage 2: Deduplication
Removes exact duplicates, false positives (occurrence-level classification), and near duplicates using TF-IDF similarity. Results in 57K unique jobs (53% reduction).

Stage 3: AI Application Extraction
Three-step LLM pipeline using O3 and GPT-5-mini models to extract, separate, and standardize AI applications from job descriptions. Performance: F1 scores improving from 0.78 → 0.84 → 0.92 across steps.

Stage 4: O*NET Similarity Analysis
Calculates similarity between extracted AI applications and O*NET occupational tasks. Filters matches at 95th percentile threshold. Uses BGE embeddings (or OpenAI) for semantic matching.

Stage 5: Exposure Calculation
Aggregates AI application usage to firm-occupation-year level. Applies two exposure methods: Hampole (share-based) and Binary. Includes intensity adjustment using $\log(1 + N_{\text{applications}})$.

Stage 6: Link to Jobs
Joins exposure measures to actual job postings in database. Includes occupational crosswalks (X28 → ISCO → O*NET). Produces diagnostics for unmatched records. Optional firm-level summary report.

Stage 7: Analysis & Reporting
Comprehensive analysis including firm-level rankings, occupation-level exposures, firm-occupation pairs, time-series trends, and sensitivity analysis. Generates visualizations, heatmaps, and summary statistics.